

Symphysis: A Trust-First, Reproducible Multi-Agent Engine for AI-Panel Expert Elicitation

Muhammad Danyal Khan^{1,*}

Juan Chiachío Ruano¹

Rahim Ali^{2,†}

¹IPM Labs, Department of Structural Mechanics, Universidad de Granada, Granada, Spain

²Department of Computer Science, Bahria University Karachi Campus, Karachi 75500, Pakistan

*Corresponding author, dan@go.ugr.es

†In memoriam

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Abstract

Expert-elicitation instruments, the Best-Worst Method (BWM), the Analytic Hierarchy Process (AHP), and hierarchical variants of BWM, are the standard tool for weighting decision criteria in specialized technical domains such as construction, wind-energy asset management, and blockchain-trust design. Assembling a sufficiently large and diverse *human* expert panel is often the actual bottleneck, not the method itself. We present Symphysis, an open-source engine that runs these instruments against a panel of individually configured AI agents, alongside or in place of a human panel. Every agent is fully specified by one portable, `did:key`-identified Agent Card, so a panel is reproducible from a single JSON file per agent rather than a set of ad hoc prompts nobody can rerun. Symphysis enforces, rather than merely documents, each agent’s declared permissions and guardrails: a schema-validation and denylist-scan pass runs on every response, every agent is sampled multiple times before a response counts toward a result, and a QA-precheck flags fabricated source claims before they enter the solved posterior. Its Bayesian hierarchical BWM solver is verified against Mohammadi and Rezaei’s published reference implementation, with one real discrepancy found and corrected in the process. A separate pooling method, HAWC-BWM, combines a human-panel posterior and an AI-panel posterior via draw-wise linear pooling across an alpha sensitivity sweep, letting the two panels’ agreement and disagreement be inspected directly rather than collapsed into a single blended number. This paper describes Symphysis’s architecture and trust design, and reports an empirical comparison of a human panel and a Symphysis AI panel run on the same BWM survey already used to calibrate the TrustRouter trust-scoring framework’s dimension weights.

Keywords: expert elicitation multi-agent systems Best-Worst Method Analytic Hierarchy Process Bayesian hierarchical modeling agent identity AI trust and safety

1 Introduction

Weighting decision criteria via structured expert elicitation is standard practice in multi-criteria decision analysis, but the practical constraint is rarely the method: it is finding enough qualified human experts, willing and able to complete a demanding elicitation instrument, in a sufficiently narrow technical specialty. This constraint is acute in domains this project family already works in: construction trust-routing decisions, wind-farm asset-management priorities, and blockchain-suitability scoring, all of which need domain-expert judgment that a generic panel cannot supply.

A growing body of work explores having large language models stand in for, or alongside, human survey respondents. Most of that work does not address two questions a reviewer of a real elicitation study will ask: can the panel be reproduced exactly, and can the agents be trusted to have answered honestly rather than guessed or fabricated a justification. We built Symphysis to answer both questions by construction rather than by convention: every agent’s identity, permissions, and behavioral guardrails are declared in one portable file and enforced in code, not left as an informal prompting convention.

1.1 Contributions

- An open-source, reproducible AI-panel expert-elicitation engine in which agent permissions and guardrails are enforced at the architecture level, not merely documented.
- A portable, `did:key`-identified Agent Card format that lets a panel be exactly respawned from a single JSON file per agent.
- A Bayesian hierarchical BWM solver verified against a published reference implementation, with one real discrepancy found and corrected during that verification.
- HAWC-BWM: a draw-wise linear-pooling method for combining a human-panel posterior and an AI-panel posterior across an alpha sensitivity sweep.
- An empirical comparison of a human panel and a Symphysis AI panel on a real elicitation instrument already used in this project family.

2 Related Work

2.1 Expert elicitation and multi-criteria decision analysis

The Best-Worst Method (Rezaei, 2015) and its Bayesian hierarchical extension (Mohammadi and Rezaei, 2020) provide the elicitation and solving methodology this system implements; the Analytic Hierarchy Process (Saaty, 1980) is the second instrument family supported.

2.2 Agent identity, permission enforcement, and prompt-injection risk

He et al. (2026, arXiv:2608.27299) show that both model-side instruction-hierarchy defenses and a common harness-side automatic permission reviewer can be defeated by untrusted tool-level content engineered to be read as a legitimate instruction rather than data. This motivates Symphysis’s deterministic, non-LLM-mediated permission check (Section 3) rather than an LLM-based intent reviewer.

2.3 Multi-agent shared-substrate risk

Papadopoulos et al. (2026, arXiv:2608.10218) show that a goal or idea can propagate agent-to-agent through ordinary communication in a shared multi-agent substrate, and that idle or under-specified agents are more susceptible while an explicit warning in an agent’s system prompt confers near-total immunity, directly relevant to any Symphysis deployment where multiple agents share one survey’s knowledge repository.

2.4 Adjacent AI-for-research-process systems

Schmidgall et al. (2026, arXiv:2608.26701) and Li et al. (2025, arXiv:2510.15624) address the related but distinct problem of AI agents accelerating or standing in for parts of the scientific research process more broadly (hypothesis generation through manuscript writing), rather than expert-elicitation surveys specifically; Symphysis is narrower in scope and, by design, sidesteps the context-window-overload and rigid-pipeline failure modes those systems’ own authors identify, since each Symphysis agent is stateless per survey round and driven by a fixed instrument rather than an adaptive long-horizon plan.

3 System Architecture

3.1 The Agent Card

Every agent is fully defined by one portable JSON file: a real `did:key` identity, role, model and hyperparameters, an optional RAG corpus, sampling policy, tool access, permissions (which data paths and providers it may use), and guardrails (denylist patterns). Given this file, plus the shared DID seed, any party can respawn an identical agent without reading Symphysis’s own source code.

3.2 Instruments

The instrument owns prompt construction and response schema; agents do not know or care which instrument they are completing. `bwm` (Best-Worst Method), `ahp` (Analytic Hierarchy Process), and `hierarchical_bwm` (several BWM comparisons in one response, for a criteria tree rather than a flat list) are implemented behind a common interface designed for further methods to be added the same way.

3.3 Enforced permissions and guardrails

A RAG-enabled agent’s declared corpus path must match one of its card’s `permissions.data_scopes` globs or the agent refuses to start; its model provider must be in `permissions.allowed_providers`. Every response is schema-validated and rejected or re-sampled on malformed output; a denylist regex scan catches prompt-injection markers or secret-shaped strings even in otherwise well-formed output; every agent is sampled multiple times, never treating a single completion as ground truth; every raw model completion, not just the parsed answer, is logged for audit.

3.4 Deterministic genuineness checks

A QA-precheck restatement is checked against ground truth, and a `sources_used` claim is checked against what was actually available in the prompt, flagging a fabricated citation before it can enter a solved result.

3.5 Solving and pooling

The classical BWM linear program (Rezaei, 2015) and the Bayesian hierarchical BWM (Mohammadi and Rezaei, 2020), verified against the original authors’ reference JAGS implementation, with one real bug found and fixed in the process, solve each panel’s responses into a posterior. HAWC-BWM combines a human-panel posterior and an AI-panel posterior via draw-wise linear pooling across a full alpha sensitivity sweep, so agreement and disagreement between the two panels can be inspected directly rather than collapsed into one blended estimate.

4 Evaluation

4.1 Design

We will run the same BWM instrument, TrustRouter’s own trust-dimension weighting survey, through both the original human expert panel and a Symphysis AI panel, solve both posteriors with the verified Bayesian hierarchical BWM solver, and pool them via HAWC-BWM’s draw-wise linear pooling across an alpha sensitivity sweep.

4.2 Expected outcome

We expect broad directional agreement between the two panels’ solved weights, with disagreement concentrated in criteria where domain-specific tacit knowledge (not captured in any agent’s RAG corpus) matters most, and we expect the enforced-permission and guardrail pipeline to produce an auditable, fully reproducible trail for every AI-panel response.

4.3 Results

5 Discussion

6 Limitations

7 Conclusion

CRedit authorship contribution statement

Muhammad Danyal Khan: Conceptualization, Methodology, Software, Investigation, Writing – original draft, Writing – review & editing. **Juan Chiachío Ruano:** Conceptualization, Supervision, Writing – review & editing. **Rahim Ali:** Conceptualization.

Data availability

The Symphysis codebase, Agent Card schemas, and survey configurations used in this paper will be made available in the project’s repository upon publication.

Competing interests

The authors declare no competing interests.

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